

Mini-Test
Modern Physics

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Text and solution for Monday's mini-test.

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WEEK IX

- A hydrogen atom is in the ground state ($1s$) in a magnetic field $B = 3\text{ T}$. Due to the *spin Zeeman effect*, the energy of the electron is shifted according to the potential $V_B = -g_s\mu_B m_s B$.
 - List the two allowed values of the spin quantum number m_s . (0.5pt each)
 - Calculate the energy splitting (absolute value) $|\Delta E|$ due to the potential between the two spin states. (1pt)

The s orbitals have $\ell = 0$, so the orbital angular momentum plays no role here.

Use: $g_s = 2$ (spin g -factor) and $\mu_B = 5.788 \times 10^{-5}\text{ eV/T}$.

Solution

The two allowed values of the spin quantum number are $m_s = \{+\frac{1}{2}, -\frac{1}{2}\}$, corresponding to $|\uparrow\rangle$ and $|\downarrow\rangle$, respectively.

In the magnetic field, the energy of each spin state is given by

$$\begin{aligned} E_{\uparrow} &= E_1 - \mu_B B \\ E_{\downarrow} &= E_1 + \mu_B B, \end{aligned}$$

where $E_1 = -E_0 = -13.6\text{ eV}$ is the energy of the $1s$ state in zero magnetic field. Note how the energy of the state with spin projection *aligned* to the magnetic field is *reduced* by the potential, meaning that that state is energetically favored.

Finally, the energy splitting is given by

$$\Delta E = E_{\downarrow} - E_{\uparrow} = 2\mu_B B = 3.47 \times 10^{-4}\text{ eV}.$$